

Geopolitical Oil Shocks, Sectoral Heterogeneity, and Aggregation Masking: Evidence from a Multi-Event Framework

Aggregate indices conceal ~95% of sectoral disruption — oil sensitivity predicts which sectors win and lose

GEOPOLITICAL OIL SHOCK



+42.3%

USO (Oil ETF)

Oil prices surged

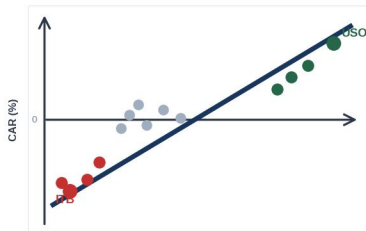
Strait of Hormuz 2026

~21% of global oil consumption transits Hormuz

Validated Across 5 Events

- | | |
|-----------------------|--------|
| ● Hormuz 2026 | Supply |
| ● Russia–Ukraine 2022 | Supply |
| ● OPEC+ Cut 2022 | Supply |
| ● Middle East 2023 | Supply |
| ● COVID-19 2020 | Demand |

SECTORAL DIVERGENCE



$$\gamma_1 = +20.18^{***}$$

1 s.d. oil sensitivity predicts
18–21 pp higher CARs

$R^2 = 0.42$ | Bootstrap CI: [15.3, 31.6]

Builds over time: +4.5 (t+1) → +25.1 (t+15)

34 US sector ETFs | 11 GICS sectors + 23 sub-sectors

AGGREGATION MASKING

59 pp sectoral range

+42.3%

-16.6%

-1.81%

S&P 500



Novel Masking Index

$$\Psi = 0.947$$

~95% concealed

5%

Crisis: 0.947 vs Normal: 0.851

Masking is worst when
accuracy matters most

72 placebo tests | 5 events

IMPLICATIONS

Supply Shocks

$\gamma_1 > 0$ (4 events)

Demand Shocks

$\gamma_1 < 0$ (COVID reversal)

$$\beta_{oil} \times \text{Supply} = +58.79^{***}$$

t = 5.22, strongest test

Who Is Affected?

Policymakers

Index understates disruption

Portfolio Managers

Benchmarks mislead

Risk Managers

Index volatility uninformative

Oil sensitivity (β_{oil}) predicts sectoral winners and losers across geopolitical shocks,
but cap-weighted indices conceal ~95% of this heterogeneity ($\Psi = 0.947$)